

Tutorial: Femoral Nerve IFC Simulation with PyFibers + COMSOL

This tutorial walks you through using this project end-to-end: from a COMSOL field export, to a running PyFibers/NEURON simulation of a femoral nerve fiber under interferential current (IFC) stimulation, to interpreting the results.

1. What you'll build

A pipeline that answers: *“For a given femoral nerve fiber, and a given IFC electrode configuration (two channels at slightly different carrier frequencies), does the fiber fire an action potential — and when?”*

COMSOL (2 unit-current field exports)



load_comsol_potentials.py → interpolate fields onto fiber coordinates



run_simulation.py → build fiber, apply IFC waveforms, run NEURON



"Action potential detected: True/False, at t = ... ms"

2. One-time environment setup (WSL)

NEURON (PyFibers' simulation engine) has no Windows pip wheels, so we use WSL Ubuntu.

In Windows PowerShell (as admin), one-time:

```
wsl --install
```

reboot if prompted, then open Ubuntu

Inside Ubuntu:

```
sudo apt-get update
```

```
sudo apt-get install -y python3-venv python3-pip build-essential
```

```
python3 -m venv ~/pyfibers_env
```

```
source ~/pyfibers_env/bin/activate
```

```
pip install pyfibers numpy scipy pandas matplotlib neuron
```

Compile the NEURON mechanism files bundled with PyFibers (one-time):

```
python -c "import pyfibers; print(pyfibers.MOD_dir)"
```

```
cd <path printed above>
```

```
nrnivmodl
```

Verify it worked:

```
python -c "import neuron, pyfibers; print(neuron.__version__, 'pyfibers ok')"
```

```
# -> 9.0.1 pyfibers ok
```

Tip: your project folder on the Windows side (e.g. C:\Users\you\...\PyFibers_ICs) is reachable from WSL at /mnt/c/Users/you/.../PyFibers_ICs.

3. Step 1 — Get your COMSOL field exports

You need **two** files, one per IFC electrode-pair channel, each containing the **potential per unit injected current** (a “lead field”). See [documentation.pdf](#) §6 / the project README for the full COMSOL recipe; in short, for each channel:

1. Build the femoral nerve tissue geometry with realistic conductivities.
2. Set one electrode pair as Terminal (current = 1 A) and Ground.
3. Run a **stationary** study (Electric Currents physics — quasi-static is fine for IFC frequencies).
4. Export x , y , z , V (μm , V) over the region your fiber passes through.

Save the two files as, e.g.:

```
comsol_exports/channel1_field.csv
comsol_exports/channel2_field.csv
```

Each row should look like:

```
1023.4, -540.2, 12500.0, 0.000182
```

(x , y , z in μm , V in volts per 1 A injected).

Don’t have COMSOL data yet? You can still test the pipeline with a synthetic point-source field — see §7 (Smoke test).

4. Step 2 — Understand the fiber model

The pipeline builds a **femoral nerve fiber** using PyFibers’ MRG (McIntyre–Richardson–Grill) double-cable model — a standard model for large myelinated peripheral axons.

Key parameters you’ll choose:

Parameter	Meaning	Typical femoral nerve values
--fiber-diameter	Axon diameter (μm)	8–16 μm for large motor/sensory fibers
--n-nodes	Number of nodes of Ranvier	More nodes = longer fiber; 51 is a reasonable default
--fiber-model	PyFibers FiberModel	MRG_INTERPOLATION (default) for myelinated fibers

The fiber is placed along the z -axis starting at the origin by default. **Your COMSOL coordinates must use the same frame** (or you should translate/rotate the fiber via `fiber.set_xyz(...)` to match the COMSOL geometry — see §8 for customization).

5. Step 3 — Understand the IFC stimulus

Two channels, each a sinusoidal current at a slightly different frequency:

Parameter	Meaning	Typical IFC values
--f1, --f2	Channel carrier frequencies (Hz)	e.g. 4000, 4100
--amplitude1, --amplitude2	Scale factors applied to each channel's unit-current field	depends on your COMSOL normalization (see §6 of the requirements doc)
--tstop	Total simulated time (ms)	enough to cover several beat cycles, e.g. 50 ms
--dt	NEURON integration time step (ms)	0.001–0.005 ms

The **beat frequency** ($|f_2 - f_1|$, e.g. 100 Hz) is computed automatically and printed — this is the low-frequency envelope that emerges where the two fields overlap.

6. Step 4 — Run the simulation

From the project root (in WSL, with the venv activated):

```
source ~/pyfibers_env/bin/activate
cd /mnt/c/Users/<you>/.../PyFibers_ICs

python src/run_simulation.py \
    --channel1-file comsol_exports/channel1_field.csv \
    --channel2-file comsol_exports/channel2_field.csv \
    --fiber-diameter 10 --n-nodes 51 \
    --f1 4000 --f2 4100 \
    --amplitude1 1.0 --amplitude2 1.0 \
    --tstop 50 --dt 0.001
```

Expected output

```
Built fiber: MRG_INTERPOLATION, 221 coordinates
Beat frequency at target tissue: 100.0 Hz
Action potential detected: True (count=1.0), at t = 1.25 ms
```

- **Built fiber: ... N coordinates** — confirms the fiber model was constructed; N depends on --n-nodes and the model's internal segmentation.
- **Beat frequency** — sanity check that your f1/f2 give the IFC envelope you intended.
- **Action potential detected** — True/False and, if True, the time (ms) the AP was detected at the recording location (default 90% along the fiber).

7. Smoke test (no COMSOL data needed)

To confirm your environment works before you have real COMSOL exports, generate a synthetic point-source field:

```
python - <<'EOF'
import numpy as np
xs = np.linspace(-2000, 2000, 9)
ys = np.linspace(-2000, 2000, 9)
```

```

zs = np.linspace(0, 50000, 50)
X, Y, Z = np.meshgrid(xs, ys, zs, indexing='ij')
pts = np.column_stack([X.ravel(), Y.ravel(), Z.ravel()])
for name, center in [('channel1_field.csv', np.array([1000, 0, 25000])),
                     ('channel2_field.csv', np.array([-1000, 0, 25000]))]:
    r = np.linalg.norm(pts - center, axis=1) + 1.0
    V = 1.0 / (4*np.pi*0.3*r) # point-source potential per unit current
    np.savetxt(f'comsol_exports/{name}', np.column_stack([pts, V]), delimiter=',')
print("synthetic exports written")
EOF

```

```

python src/run_simulation.py \
  --channel1-file comsol_exports/channel1_field.csv \
  --channel2-file comsol_exports/channel2_field.csv \
  --fiber-diameter 10 --n-nodes 21 \
  --f1 4000 --f2 4100 --amplitude1 50 --amplitude2 50 \
  --tstop 10 --dt 0.005

```

If this prints Built fiber: ..., Beat frequency: 100.0 Hz, and an AP detection line without errors, your environment is correctly set up.

8. Customizing the pipeline

Sweeping stimulation amplitude (finding a threshold)

To find the minimum amplitude that triggers an AP, run the simulation in a loop with increasing `--amplitude1/--amplitude2` (keeping their ratio fixed) until Action potential detected: True appears. (PyFibers also has a built-in `ScaledStim.find_threshold` helper — see the PyFibers docs — which could replace this manual sweep in a future extension.)

Different fiber models or sizes

Change `--fiber-model` to any `pyfibers.FiberModel` member (e.g. `MRG_DISCRETE`, `SUNDT`, `RATTAY` for unmyelinated fibers) and adjust `--fiber-diameter` / `--n-nodes` accordingly. Smaller-diameter / unmyelinated models will generally need smaller `--dt`.

Aligning fiber and COMSOL coordinate frames

If your COMSOL geometry doesn't place the femoral nerve fiber along the z-axis from the origin, translate the fiber to match before assigning potentials:

```

fiber.set_xyz(x=..., y=..., z=...)
v1, v2 = load_channel_potentials(..., fiber.coordinates)

```

(Add this call in `run_simulation.py` between fiber construction and potential loading.)

Three IFC channels / “premodulated” IFC

The pipeline currently supports exactly two channels (`fiber.potentials = [v1, v2]`, two waveforms, two stimamps). To add a third (e.g. for “interferential” current with a premodulated channel, or a tripolar electrode setup), extend each list to three elements and provide a third COMSOL export.

9. Troubleshooting

Symptom	Likely cause	Fix
ModuleNotFoundError: No module named 'neuron'	Running outside the WSL venv	source ~/py-fibers_env/bin/activate
ValueError: argument not a density mechanism name	NEURON .mod mechanisms not compiled	Run nrnivmodl in pyfibers's MOD directory (\$2)
Interpolated potentials are all NaN / pipeline crashes	Fiber coordinates fall completely outside the COMSOL export's spatial range	Check units (μm vs m) and coordinate frame alignment between fiber and COMSOL
Action potential detected: False even at high amplitude	Amplitude too low for your COMSOL normalization, or fiber not near the field maxima	Increase --amplitude1/2, check fiber.coordinates lie within the high-field region of your COMSOL model
Beat frequency looks wrong	--f1/--f2 swapped or equal	beat_frequency = f2 - f1 ; ensure they differ as intended

10. Where to go next

- [README.md](#) — quick reference / setup commands
- [documentation.pdf](#) — full requirements reference and line-by-line code walkthrough
- [PyFibers documentation](#) — fiber models, ScaledStim options, threshold-finding utilities